What Matters in Frozen Self-Supervised Representations for Low-Resource ASR: A Controlled Study of Layer and Decoder Choice

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Abstract

Self-supervised speech models such as wav2vec 2.0 are widely used as *frozen* feature extractors for low-resource ASR, yet several practical choices remain under-documented: which encoder layer to tap, how complex the downstream decoder should be, and how much labelled data is needed. We present a controlled study on LibriSpeech that isolates these three axes under a single fixed-budget setup with a frozen wav2vec 2.0 encoder and a character-level CTC head. We find that (i) the representation layer is the single largest lever—a middle layer (9) reduces WER from 0.86 (final layer) to 0.30—because the pretraining-only encoder's top layers specialise to the pretext task; (ii) a simple non-linear per-frame head is sufficient: a two-layer MLP matches or beats a Transformer decoder with 4.3× the parameters, indicating that cross-frame modelling is largely redundant with the encoder's own context; and (iii) inference latency is dominated by the encoder forward pass, so decoder size (a 77× range) barely changes RTF. Data scaling exhibits clear diminishing returns, suggesting that *adapting* the representation rather than adding data is the more promising direction.
Index Terms—self-supervised learning, wav2vec 2.0, low-resource ASR, CTC, representation probing.

1 Introduction

Self-supervised learning (SSL) models—wav2vec 2.0 [1], HuBERT [2], WavLM [3]—are pretrained on large amounts of unlabelled speech and yield general-purpose representations that transfer well to downstream tasks. A dominant recipe for low-resource ASR is to *freeze* the SSL encoder and train only a lightweight downstream head on top, a form of representation probing that avoids overfitting the millions of encoder parameters on scarce labelled data.

While this recipe is common, three design decisions are usually made by convention rather than measurement: (1) *which* of the encoder's many Transformer layers to use, (2) how *complex* the downstream decoder should be, and (3) how much labelled data is actually required. These are rarely answered together under one controlled setting. In this paper we hold a single baseline system fixed and vary exactly one factor at a time, on LibriSpeech, with a frozen wav2vec 2.0 encoder and a character-level CTC head.

Our contributions are: (i) a controlled ablation across representation layer, decoder structure and data size that ranks their importance for frozen-SSL low-resource ASR; (ii) the observation that a simple non-linear per-frame head suffices, while added cross-frame capacity is redundant with the encoder's own contextualisation; and (iii) the finding that, under a frozen encoder, inference latency is set by the encoder and is essentially independent of decoder size, so accuracy and decoder complexity are decoupled from speed.

2 Method

2.1 Frozen SSL encoder

We use `wav2vec2-base` [1], kept frozen throughout. A seven-layer convolutional feature encoder downsamples the 16 kHz waveform by a factor of 320, producing one 512-d frame every 20 ms (50 frames/s). A 12-layer Transformer then produces contextual hidden states. We expose the hidden states of a chosen layer l and treat them as fixed features.

2.2 Representation and CTC head

The selected hidden states $C \in \mathbb{R}^{T \times 768}$ are passed to a downstream decoder that emits, for every frame, a distribution over a 30-token character vocabulary (blank, unknown, space, apostrophe and the 26 letters). The model is trained with the CTC objective [4], which marginalises over all frame-to-character alignments and requires no frame-level supervision. At inference we use greedy decoding: per-frame $\arg\max$, followed by collapsing repeats and removing blanks. The full pipeline is shown in Fig. 1.

2.3 Decoder variants

We compare three heads that all preserve the time axis (so CTC alignment is unaffected): a **linear** head (a single projection; a pure linear probe), an **MLP** head (LayerNorm, a hidden layer with GELU, then a projection; non-linear but still per-frame), and a **Transformer** head (an input projection, sinusoidal positional encoding and two self-attention layers; the only variant that mixes information across frames).

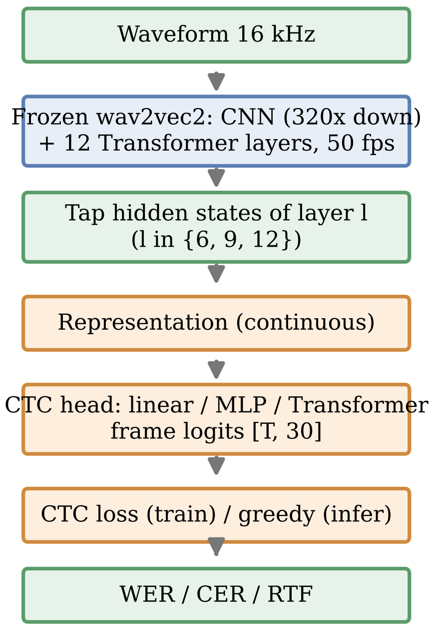

Figure 1: System pipeline. The SSL encoder (blue) is frozen; only the representation and CTC head (orange) are trained. Our three ablation axes are the tapped layer l, the decoder structure, and the amount of training data.

3 Experimental Setup

Experiments use LibriSpeech (clean) [6] in streaming mode, with 3600 training and 500 validation utterances at 16 kHz for the baseline. All systems are trained under a single *fixed budget*: 10 epochs, batch size 8, learning rate 10^{-3} with a cosine schedule and early stopping, and a single random seed. We therefore report a fixed-budget comparison rather than converged numbers; this is revisited in Section 5. Unless noted, the baseline is { wav2vec2-base frozen, layer 9, MLP head, 3600 utterances }, and each study perturbs exactly one factor. We report Word and Character Error Rate (WER/CER), the Real-Time Factor (RTF), and the substitution/deletion/insertion counts for error analysis.

4 Results and Analysis

4.1 Which layer? (Representation layer)

Table 1 and Fig. 2 vary the tapped layer with the head, data and budget fixed. A middle layer (9) is clearly best, while the final layer (12) is dramatically worst. This reproduces the well-known acoustic→linguistic→pretext trajectory of SSL representations [5]: because `wav2vec2-base` is pretrained-only (never fine-tuned for ASR), its top layers specialise back to the contrastive/quantisation pretext objective and lose directly usable content information. Simply changing the

Table 1: Layer ablation (MLP head, 3600 utterances, 10 epochs). S/D/I = substitutions/deletions/insertions.

Layer	WER	CER	Val. loss	S/D/I
6	0.4138	0.1186	0.4604	3334/301/300
9	**0.2954**	**0.0768**	**0.3088**	2459/238/112
12 (final)	0.8596	0.3426	1.1925	6150/1837/188

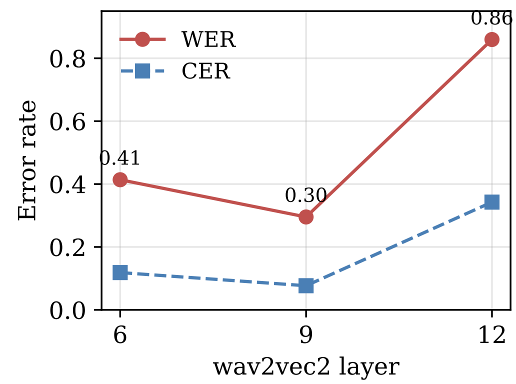

Figure 2: WER and CER as a function of the tapped wav2vec2 layer. The middle layer is best; the final layer collapses.

tapped layer lowers WER from 0.86 to 0.30 (a ∼66% relative reduction) at *zero* additional parameters or training—the largest single lever in this study.

4.2 How complex a decoder? (Decoder structure)

Table 2 and Fig. 3 vary the head at the best layer (9). Two effects stand out. First, *non-linearity is the main lever*: moving from the linear probe to the MLP cuts WER from 0.60 to 0.30 (∼51% relative), showing that the layer-9 features are not fully linearly separable but a single non-linear per-frame mapping recovers most of the gap. Second, *cross-frame modelling does not help*: the Transformer head (1.78M parameters) is slightly worse than the MLP (0.41M). Two factors explain this: (a) wav2vec 2.0 is itself a 12-layer Transformer, so its layer-9 features are already strongly contextualised and an extra Transformer is redundant; and (b) under the fixed 10-epoch budget the larger head is under-optimised—its validation loss (0.3922) is *higher* than the MLP's (0.3088), which indicates under-fitting rather than over-fitting. We therefore state only that cross-frame capacity gives no gain *under this budget*, without claiming it is fundamentally inferior.

Finally, RTF is essentially constant (0.010–0.011) across a 77× range of decoder parameters. Because the frozen encoder forward pass dominates computation, decoder complexity is decoupled from inference speed: one can choose the head on accuracy grounds alone.

Table 2: Decoder ablation (layer 9, 3600 utterances, 10 epochs).

Decoder	Params	WER	CER	Val. loss	RTF
Linear	23,070	0.6027	0.1846	0.6456	0.0102
MLP	410,654	**0.2954**	**0.0768**	**0.3088**	0.0111
Transformer	1,784,094	0.3163	0.0845	0.3922	0.0104

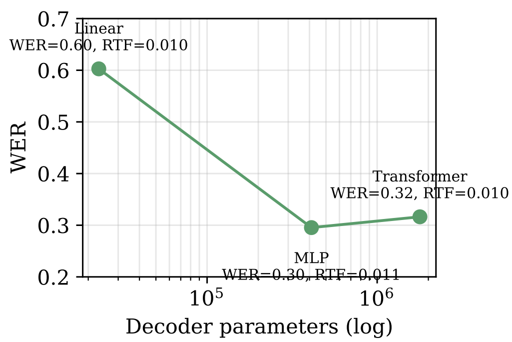

Figure 3: WER versus decoder parameters (log scale). A non-linear MLP is best; the much larger Transformer does not improve accuracy, and RTF is essentially unchanged.

Table 3: Training-set size (layer 9, MLP head, 10 epochs).

Utterances	900	1800	3600	5400	7200
WER	0.4278	0.3449	0.2954	0.2579	0.2493
CER	0.1174	0.0911	0.0768	0.0674	0.0634

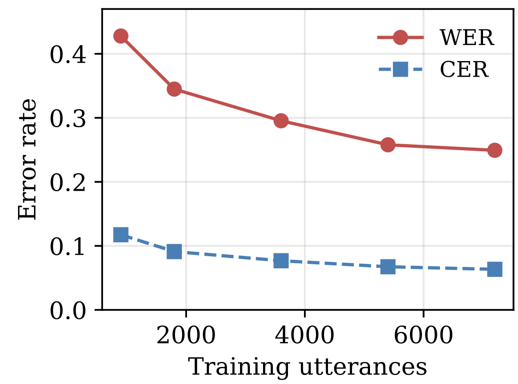

Figure 4: WER and CER versus training-set size. Errors fall with clear diminishing returns.

4.3 How much data? (Training-set size)

Table 3 and Fig. 4 scale the training set from 900 to 7200 utterances (layer 9, MLP). WER decreases monotonically but with clear diminishing returns: doubling from 900 to 1800 helps substantially (0.43 $\rightarrow$ 0.34), whereas doubling from 3600 to 7200 yields only 0.30 $\rightarrow$ 0.25 ($\sim$16% relative). The flattening curve suggests that a frozen representation with a lightweight head is approaching a ceiling, and that further gains are more likely to come from *adapting* the representation (e.g. parameter-efficient fine-tuning [7]) than from additional labelled data.

4.4 Error analysis

Across configurations, healthy systems (layer 9, MLP, larger data) are dominated by *substitutions* with few deletions. In contrast, the two failing configurations—the final layer and the linear head—show sharply elevated *deletions* (1837 and 828 respectively, versus $\sim$238 for the baseline out of 9510 reference words). This is the signature of a blank-dominated CTC output collapsing toward empty transcriptions, and it matches the large CER/WER gap of the final layer (CER 0.34 but WER 0.86): characters are partially correct but almost no whole word is, indicating an under-optimised or mismatched representation rather than random output.

4.5 Synthesis

Anchored at layer 9, the three axes can be ranked by leverage: *representation layer* $\gg$ *per-frame non-linearity* $>$ *data size* $>$ *cross-frame capacity* (≈ 0). In short, for frozen-SSL low-resource ASR, what matters is choosing the right layer and a simple non-linear head; extra decoder capacity and additional data both give limited marginal returns.

5 Discussion and Limitations

All results use a fixed 10-epoch budget and a single seed, so they should be read as a fixed-budget comparison; larger heads such as the Transformer may be under-trained (as their higher validation loss suggests), and longer training or multiple seeds would tighten the close comparisons. As preliminary work toward discrete units, we also implemented a k-means discretisation path (continuous features $\rightarrow$ cluster id $\rightarrow$ trainable embedding $\rightarrow$ CTC) that reports token rate, bitrate and codebook utilisation; however, under the same tight budget the from-scratch unit embedding did not train well, so we do not report discrete ASR accuracy as a reliable result and leave it to future work. Promising directions include parameter-efficient adaptation of the encoder (e.g. LoRA) to break the data ceiling, feature caching to afford longer training, and cross-model comparison with HuBERT and WavLM.

6 Conclusion

We presented a controlled study of frozen wav2vec 2.0 features for low-resource CTC ASR. The tapped layer is the dominant factor—a middle layer far outperforms the pretraining-specialised final layer—while a simple non-linear per-frame head is sufficient and added cross-frame modelling is redundant with the encoder's own context. Inference latency is set by the frozen encoder, and data scaling shows diminishing returns. The practical recommendation is to spend effort on layer selection and a lightweight non-linear head, and to reserve compute for adapting the representation rather than enlarging the decoder or the dataset.

References

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